DeltaSieve: Targeted Visual Validation of Spreadsheet Transformations

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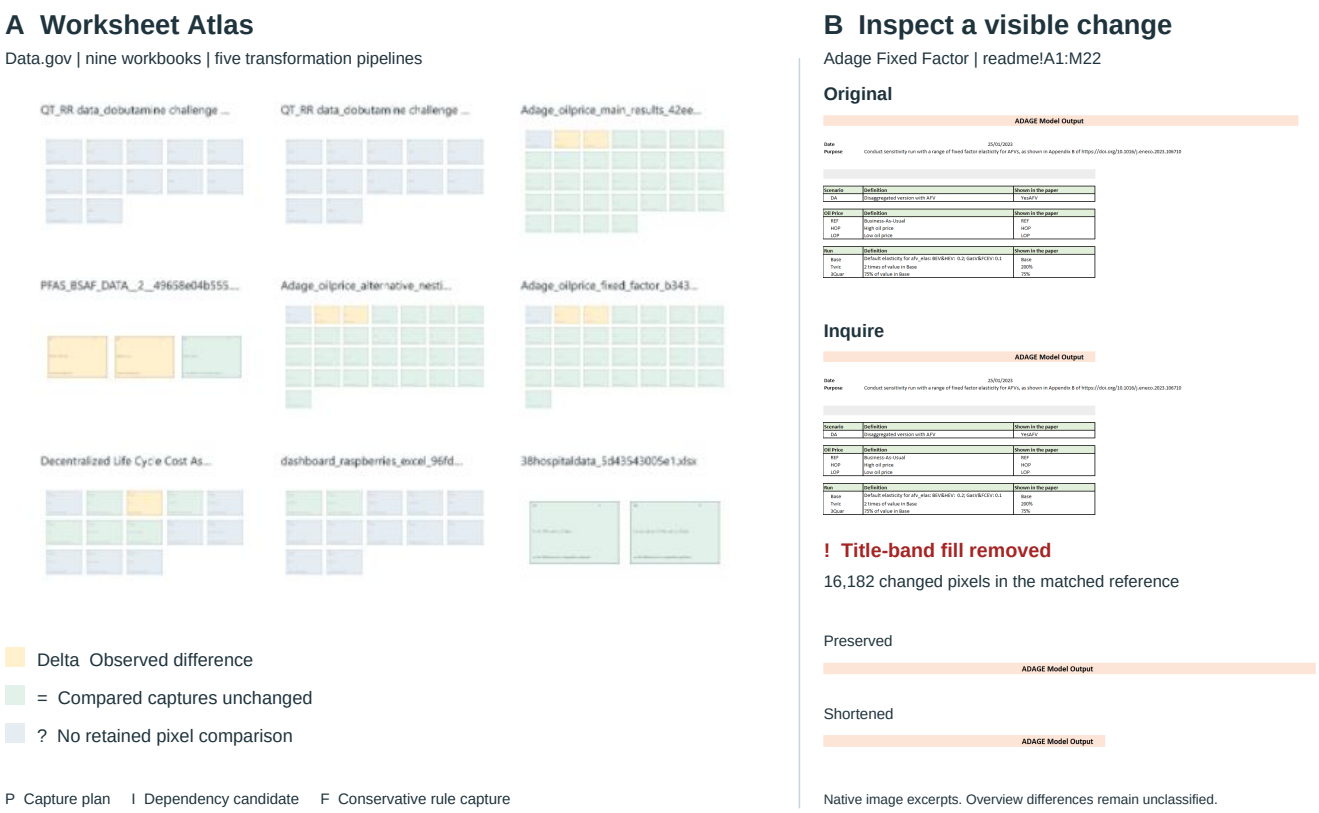


Fig. 1. From worksheet-level evidence to native image inspection. The left view shows nine Data.gov workbooks and their 144 known sheets. Amber marks recorded differences with unclassified cause, green denotes unchanged compared captures, and blue-gray retains missing pixel coverage. Planning badges retain recorded decisions independently of the outcome. The right view enlarges a verified Adage title-band change in Inquire. The matched reference has 16,182 changed pixels. The displayed extent also remains shortened in Check Performance and NXPowerLite, while SheetShrink and XLStylesTool preserve it. These grouped observations concern this reference range. Interactive Atlas snapshot.

***Abstract*—Checking whether a spreadsheet transformation changes its appearance can require rendering large worksheets in Excel. DeltaSieve uses differences in worksheet XML to select regions for native image comparison. It resolves cell values, inherited formatting, row and column geometry, and dependencies of conditional formatting and charts. Unresolved dependencies trigger broader captures. The selected regions are rendered in Excel and compared pixel by pixel. Separate opening, saving, and rebuilding controls record changes introduced by Excel during preparation. The comparison study is organized around 49 workbooks from Data.gov, Destatis, and Enron and five transformations produced by SheetShrink, NXPowerLite, XLStylesTool, Check Performance, and Inquire. Controlled examples show how removing a cell fill or an inherited border changes the rendered worksheet. Reports connect findings to worksheet coordinates, paired images, capture geometry, and execution measurements.**

I. Introduction

A spreadsheet transformation can preserve cell values and still change the visible document. Removing an empty formatted cell can expose a background or alter automatic row height. Changing a referenced input can affect a distant conditional-formatting region or chart. Checking only edited cell payloads therefore leaves important transformation effects unobserved.

Native rendering supplies an operational way to observe these effects in the spreadsheet application that users employ. Exhaustive rendering, however, can be expensive for large sheets and corpora. Stored worksheet extents may also include empty or uniformly formatted tails. Dividing these extents by a fixed cell grid can create many images despite very little finite visible content. The challenge is to select useful rendering work while retaining explicit uncertainty about unsupported features and incomplete comparisons.

Figure 2 illustrates a tested conditional-formatting change retained by conservative capture. DeltaSieve combines package-level change analysis with native visual comparison. It follows effective style identities across table renumbering, considers layout and dependency effects, and records the reasons for its capture choices. Supported conditional-formatting and chart dependencies can establish that an unchanged input cannot have changed the associated presentation under the supported model. Changed or unresolved dependencies remain capture candidates. The native stage measures geometry, renders bounded image batches, and compares retained evidence under a recorded state-pair policy.

A dependency-aware planner is useful only if its reduced work retains the relevant effects in the evaluated domain. We therefore separate selector coverage, pixel observation and defect attribution. Independent full-reference samples and mutations test detection. Repeated same-input observations test rendering variability. A changed image remains an observation whose cause may require further evidence.

Existing spreadsheet dependency slicing motivates careful treatment of indirect effects [1]. Existing workbook comparison tools also provide reports of structural and cell-level changes [2]. An interactive Observatory and portable report make those observations inspectable.

DeltaSieve grew out of the need to investigate visual regressions in SheetShrink, a workbook cleanup system described in a companion manuscript [3]. That manuscript studies cleanup rules and preservation structures. Here we study the selection, capture and evidence needed to evaluate rendered changes across workbook transformations, including alternative cleanup tools. The companion working PDF is included in the supplementary material at related-manuscripts/sheetshrink.pdf.

II. Preservation and Comparison Contract

A. Inputs, states, and observations

DeltaSieve validates a pair of spreadsheet states produced by a transformation. The package representation supplies evidence about where visible changes could occur. Native Excel rendering supplies the image observations for selected regions. A candidate is a reason to inspect a region; pixel inequality is an observation that still needs attribution.

The fast comparison uses Original and transformed workbooks after a full rebuild and save in their existing, separately owned Excel processes. This gives both partners the native recalculation step. Stored and ordinary-saved XML controls remain available for follow-up. Their changes explain potential causes without automatically enlarging the rebuilt-pair capture plan. An independently requested Check Performance comparison has its own transformed state and coverage.

B. Relevant visual area and candidate coverage

The relevant area includes values, formulas, effective visible formatting, text overflow, and objects. A union across relevant input states is needed to cover content removed by a transformation. Empty trailing geometry has a separate contract. The capture plan must retain enough evidence to distinguish a change inside the relevant area from a larger or smaller exterior canvas.

Candidate selection considers changed cell properties and layout, as well as regions affected through conditional formatting and graphical dependencies. Style and shared-string references require comparison of their effective contents because saving can renumber the corresponding tables. Supported reference reasoning can exclude a dependent region when its rule and inputs are unchanged. A changed dependency makes its affected region a candidate. Unresolved reference semantics require explicit fallback coverage.

A selected-region comparison does not by itself demonstrate that every relevant change in the workbook was selected. We evaluate that property using independently rendered reference areas and controlled mutations. The evaluation must record which reference cases completed and which exceeded their budgets.

C. Verdicts, failures, and reusable evidence

A successful comparison requires usable evidence from both partners. Missing hashes, failed native commands, and timeouts remain unsuccessful outcomes.

The raw pixel and geometry observations remain attached to a finding after review. Product regressions, rendering variation, environment-dependent values, and verified capture-extent differences have separate classifications. A volatile formula path explains a possible cell-value change while other changed pixels in the same image can require further analysis. No global pixel tolerance is assumed.

Reusable observations bind the input states, relevant configuration, producing validation implementation, and declared rendering environment. Package-content identity is evaluated independently of ZIP entry order and compression. Reusing a completed study preserves its source manifest and compatibility checks. Portable reports retain the observations and their provenance for inspection without rerunning Excel.

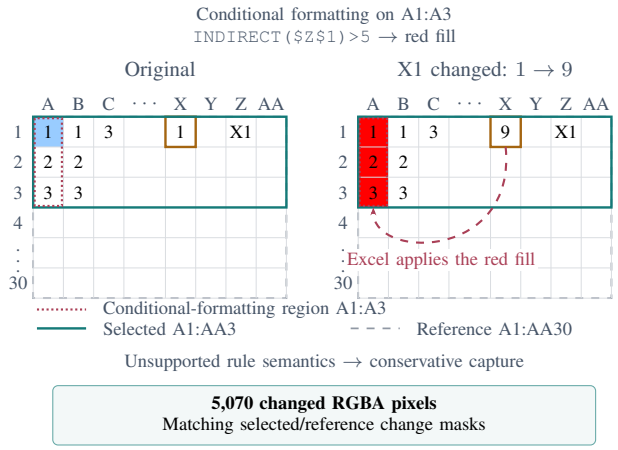


Fig. 2. Schematic reconstruction of a native-tested synthetic control [4]. X1 changes from 1 to 9 while Z1 retains "X1" and the conditional rule remains unchanged. The rule turns A1:A3 red. The tested selector uses conservative capture of A1:AA3 because the rule is unsupported. Its change mask matches the independent A1:AA30 reference. Intermediate columns and empty rows are compressed.

III. Method

A. Comparison subject and observation model

Figure 2 relates changed input, selected native coverage and retained evidence. The input is an original/transformed pair, a state-preparation policy and a renderer environment. The principal comparison opens separate owned Excel processes and observes both partners after full rebuilding in place. Ordinary-save and pre-rebuild controls retain the effects of serialization and calculation. Conversion of a legacy XLS source precedes this subject and has its own failure outcome.

Let A and B denote the compared states, E their justified relevant visual area and C the selected observation regions. The intended selector obligation is to include every transformation-induced visual difference inside E in C. Pixel equality on C tests the selected observations. It does not establish that the selector met that obligation. Independent reference captures and visible mutations evaluate this second question. Renderer variability and unsupported dependencies remain explicit boundaries of the model.

B. State preparation and attribution controls

Opening a workbook can repair or normalize its stored representation, saving can remove structure, and recalculation can change cached results. The comparison therefore names the operations that produced each observed state. Let W_0 be the raw package, W_s its ordinary-save control and W_r the rebuilt saved control. The transformed workbook has corresponding controls. Separately prepared Check Performance outputs enter as transformed inputs.

Control differences explain changes introduced by opening, saving or rebuilding. Legacy format conversion and explicit repair produce separately identified artifacts.

Each control records its parent identity, native environment and generation recipe. A reusable control artifact saves preparation work but supplies no new visual verdict. If opening or a required operation fails, that state remains unavailable. This separation also permits attributing a reduced package to Excel's save behavior without crediting that reduction to the tested cleanup operation.

C. Streaming change evidence

The package analyzer uses borrowed XML byte views with bounded hierarchy and attribute state. It resolves shared strings, cell formats and relationships before interpreting a coordinate change. String and formula interpretation can require decoding, but worksheet analysis does not instantiate a general XML DOM or an Excel object per cell.

Cell evidence is written as fixed-size records containing coordinates, a SHA-256 fingerprint and classification flags. The current record is 40 bytes. The record stream resides in an owned scratch file instead of a heap map of cell objects. Sorted streams are merged by coordinate, distinguishing changed, added and removed cells. Equal stream digests avoid repeated cell comparison. The merge is linear in the combined record counts, excluding XML parsing, style resolution, hashing and selected dependency work. Scratch I/O is an explicit cost of this memory organization. Shared strings, styles, layout records, selected changes and native images remain additional memory consumers.

Figure 3 gives the streaming comparison algorithm. The writer requires unique ascending numeric coordinates, rejects nested or unclosed cells, and hashes a cell's text incrementally. A very large cell body therefore need not become one temporary string. Record readers require complete records. A truncated scratch file cannot become a shorter valid sheet. This strict stream contract permits a two-way merge without a separate sorting pass for supported ordered worksheets.

The fingerprint preserves distinctions that the model has not justified merging. Unsupported serialization can therefore produce conservative extra work rather than a semantic-equivalence claim. Canonicalizing style-table references is a specific supported operation, not a general normalization of arbitrary formulas, XML extensions or renderer behavior.

A format index has no cross-package meaning. Style identity resolves fonts, fills, borders, number formats and base-format inheritance, so a permutation of format tables does not create a change at every cell. Changed shared context, including themes or unsupported style representations, retains the conservative path. A removed blank cell can expose an inherited row or column format even when values are unchanged. Conversely, an absent untyped blank and an empty untyped record can supply the same blank input to a conditional rule. The full visual fingerprint and the rule-input fingerprint therefore have different responsibilities.

Figure 4 shows the streaming XML token reader.

D. Dependency-aware candidate selection

Direct value, formula, style and layout changes create candidates. Effects propagate beyond their edited coordinates.

Require: Worksheet states A, B and resolved style contexts
Ensure: Added, removed and changed cell records
function WRITESTREAM(W)
  Create an owned scratch stream F and stream hash H
  for all XML events in W **do**
    Update the current cell's key, flags and incremental hash
    if a cell completes **then**
      Require a unique, strictly ascending coordinate
      Encode its key, flags and digest as a 40-byte record
      Append the record to F and update H
    end if
  end for
  Require no unclosed cell; flush F
  return (F, H)
end function
$(F_A, H_A) \leftarrow$ WRITESTREAM(A)
$(F_B, H_B) \leftarrow$ WRITESTREAM(B)
if $H_A = H_B$ **then**
  return no cell changes
end if
$a \leftarrow$ NEXT(F_A); $b \leftarrow$ NEXT(F_B)
while $a \neq \perp$ or $b \neq \perp$ **do**
  if $b = \perp$ or ($a \neq \perp$ and $a.key < b.key$) **then**
    Emit removed a; $a \leftarrow$ NEXT(F_A)
  else if $a = \perp$ or $b.key < a.key$ **then**
    Emit added b; $b \leftarrow$ NEXT(F_B)
  else
    if $a.digest \neq b.digest$ **then**
      Emit changed pair (a, b)
    end if
    $a \leftarrow$ NEXT(F_A); $b \leftarrow$ NEXT(F_B)
  end if
end while

Fig. 3. Streaming cell comparison. Keys are numeric $(row, column)$ pairs. Cell hashes use resolved style identities. NEXT reads one complete record, returns $\perp$ at end of stream and rejects truncated records. Nested cells and invalid coordinates also fail analysis. The merge retains only its current records; emitted changes are collected separately.

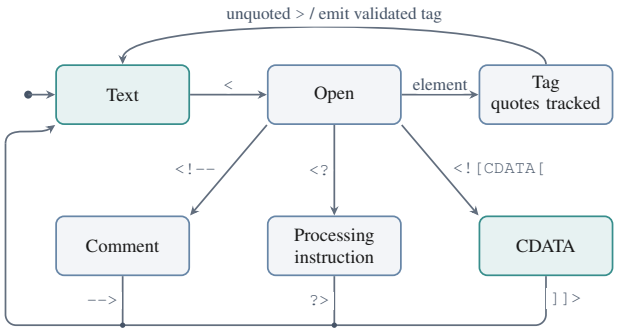


Fig. 4. DeltaSieve's six-state token reader. Quote tracking prevents > inside an attribute value from closing a tag. Complete comment, instruction and CDATA terminators return to Text along shared edges. Final acceptance requires Text state, zero open-element depth and exactly one root. Partial tokens survive read boundaries. Invalid markup and resource-limit violations return errors.

A row-height change can move later content, a column-width change can move content to its right, and a merged or centered span can extend the required text context. Changed inherited row formats are coalesced into contiguous runs, with unchanged rows splitting the runs. This reduces candidate objects while preserving the union of requested regions.

For bounded, valid, pairwise disjoint merge lists, comparison canonicalizes the rectangle order. Reordering identical rectangles therefore creates no layout candidate. Valid row `thickTop`/`thickBot` hints instead trigger native effective-height checks, including on fixed-height rows. A measured height change retains downstream coverage. Unsupported merge structures or invalid hints retain conservative fallback.

```
<!-- Only B1 changes: 0 to 1. -->
<conditionalFormatting sqref="D2:D40">
 <cfRule type="expression" dxfId="0">
  <formula>$B$1&gt;0</formula>
 </cfRule>
</conditionalFormatting>
<!-- The rule is unchanged. -->
<!-- D2:D40 still needs observation. -->
```

Fig. 5. Changing B1 can change the fill in D2:D40 while its serialized cells and rule remain unchanged. The application area becomes a native capture candidate. This example assumes a supported rule and unchanged differential style. Unknown formulas retain fallback coverage. Abbreviated XML.

Conditional-format selection resolves the supported rule formula, application area and referenced inputs. A changed precedent selects the affected application region even when its own appearance is unchanged. Local rules, area-wide inputs and explicitly referenced inputs require different dependency scopes. Exclusion requires unchanged supported rules, their relevant input fingerprints and layout. Unknown names, dynamic references, extensions or unresolved scope retain fallback coverage. DeltaSieve does not evaluate the rule to predict Excel's final color. The native renderer supplies that observation.

Figure 5 shows an unchanged rule whose precedent changes. Selection follows the dependency to its application region. It does not infer that unchanged serialized cells inside that region imply unchanged pixels.

E. Invert supported rule references

Enumerating every target cell of a whole-column conditional rule would defeat the purpose of selective rendering. For supported A1 references, DeltaSieve instead maps each changed source coordinate back to dependent target intervals. A relative scalar reference at row a in a rule anchored at row h reads source row $a + t - h$ when applied at target row t. A changed source row x therefore selects target row

$$t = x - a + h.$$

An absolute reference to row a selects all permitted target rows when $x = a$, and none otherwise. Columns follow the corresponding rule. Range endpoints can independently mix absolute and relative axes. The implementation solves the resulting axis intervals, combines row and column intervals and intersects them with the application and comparison extents. It includes both shared and per-area anchor interpretations conservatively for disjoint application areas.

This inversion avoids materializing the entire worksheet grid. Its work still depends on changed inputs, supported

TABLE I
SELECTION OBLIGATIONS. AN EXCLUSION APPLIES ONLY WITHIN THE SUPPORTED REPRESENTATION AND THE RECORDED STATE PAIR.

Effect	Required context
Cell formatting	Resolved cell, row and column components
Row/column layout	Native effective geometry and later content
Conditional rule	Rule, application region and supported input closure
Chart	Source cells, package graph, formatting and live placement
Drawing or OLE	Resolved owner, frame, ink or preview and native bounds
Uniform tail	Complete transitions and a finite relevant envelope

reference expressions and application areas. Cross-sheet references require a unique supported source sheet. Dynamic, external or ambiguous references retain fallback. Exhaustive small-grid regression controls compare the inverse result with direct forward reference evaluation, including mixed axes and reversed range endpoints. These tests verify the supported mapping, not arbitrary Excel formula semantics.

Charts have a separate dependency closure. Equal chart XML and cached series values are insufficient because source cells can change independently of the cache. The exclusion gate requires unchanged chart parts, supported source references, source values and styles, source layout and visibility, conditional dependencies, shared package context and complete equal native placement. Chart sources are inspected beyond the worksheet capture envelope. Missing partners, ambiguous names, user-shape dependencies and unsupported references retain chart capture. A conservative mode exports every chart.

Graphical objects require both package identity and native geometry. Relationship mapping binds an object to its actual worksheet. Supported lines include their stroke envelope. Zero geometric height alone cannot exclude visible ink. Cached previews of embedded objects can render even when Excel labels their shape kind unsupported. Unsupported objects therefore prevent a finite-bound claim unless their own explicit qualification succeeds.

F. Finite observation envelopes

The relevant envelope includes values, formulas, supported visible formatting, text context and qualified object extents from both compared states. Removed content therefore remains in scope. Native-qualified drawing bounds refine the rebuilt-state envelope. Save-control changes retain their own attribution records without automatically enlarging that distinct rebuilt comparison.

An empty exterior row cannot move earlier content. Effective-height selection and image selection use the same lower union boundary, while earlier empty rows remain relevant. Native zero-width and zero-height ranges keep their geometry and a non-rendered outcome. A transition to a visible range remains a finding. No white image is invented for a missing or zero-area observation.

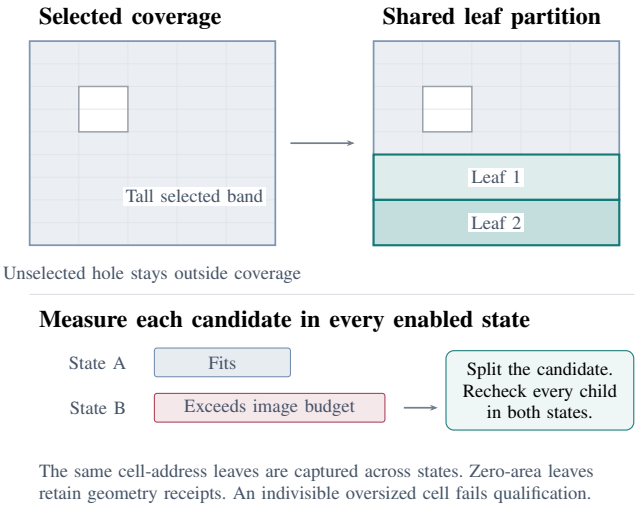


Fig. 6. Coverage-preserving adaptive capture partitioning, shown schematically. Coalescing retains holes. If a candidate exceeds the image budget in any enabled state, measured geometry drives a finer partition, which is checked again in every state. Cell-address coverage stays fixed while native image geometry can differ. Other selected regions are omitted from the lower detail.

Repeated inherited backgrounds need a finite-bound argument. For supported borderless solid fills, the analyzer retains explicit cells, row and column transitions, contextual neighbors and supported conditional application areas. Uniform unfilled column styles beneath a solid row background introduce no additional horizontal fill edge. Fonts and widths alone do not create such an edge in an empty band. Column fill transitions, inherited column borders, unknown styles and unsupported rules remain relevant. Finite row borders retain the full width through XFD, ending vertically at the last finite row transition and its following context [5]. Supported finite filters and conditional regions remain in the envelope with following context. Full-grid or unresolved application areas retain the conservative path.

These conditions permit less rendering without sampling the selected union. An unresolved outlying object can still force a full-grid plan and exhaust the available rendering budget.

G. All-state adaptive geometry

Figure 6 illustrates the coverage invariant and the effect of measuring the same candidate in multiple native states. Candidate coalescing preserves holes and worksheet identity. Independent ranges are measured in batches through the native adapter. Every live state contributes to image-size qualification. A range hidden in one state can be large in its partner, so any oversized state forces further splitting. Each child is checked again. Large hidden spans can yield zero-area leaves rather than thousands of identical image requests.

The planner respects both area and edge budgets. A conservative point-to-pixel allowance guides partitioning, but returned image dimensions are independently validated. A single cell exceeding the budget fails instead of being omitted. Measured row and column geometry handles hidden or irregular

spans that make a fixed cell-count tile unreliable. Later states can repartition an area and backfill earlier observations.

Internal boundaries retain required neighboring context for overflow, merged text and other supported layout effects. The final interior is cropped without resampling. A context region can change rendering even when its own pixels are later cropped. The renderer controls therefore compare a whole region with contextual parts and check seam consistency. Agreement of one seam is a local control, not a monolithic oracle for the workbook.

H. Native image qualification

Native images use bounded batches of Office.js `Range.getImage()` requests, synchronized before consumption.

A successful API response does not prove complete capture. The worker verifies expected dimensions, budget compliance and crop geometry before accepting the image. Returned PNG bytes can first be hashed to avoid uploading identical payloads. Unequal encodings still require decoded-pixel comparison, because PNG serialization is not visual identity. Exact canonical RGBA pixels supply the comparison, with no registration, resampling or general color tolerance. Image receipts bind command, region, state and slot identity, preventing reordered or incomplete upload batches from becoming comparisons of unrelated images.

Raw geometry and origins remain attached to the observation. An exterior-size classification can admit an unpaired right or bottom extension only when each pixel exactly repeats the common boundary. A zero-area counterpart requires an entirely opaque-white retained extent for the corresponding empty-extent classification. These derived classifications retain the original observations. They do not erase changed overlapping content or establish a global tolerance.

I. Attribution after a finding

A native difference can originate in the transformation, calculation, capture context or the environment. DeltaSieve keeps the observation while diagnostics examine those hypotheses. The formula tracer follows supported A1 references, absolute axes, quoted sheet names, ranges and translated shared formulas from changed saved values. It searches for paths to random/time or environment-dependent functions. Cycles, unsupported expressions and bounded-search exhaustion remain explicit. The search budget limits explanation work, not capture coverage.

A path to a volatile function establishes a dependency, not exclusive causation. Other changed pixels in the same region can have another cause. Source cached values also cannot stand in for values saved from the compared rebuilt states. Repeated original/original controls and unchanged-package controls test renderer behavior independently of the transformation. Counterfactual restoration of a suspected removed witness tests a narrower causal mechanism. None of these operations increases pixel tolerance or automatically clears an unrelated finding.

J. Omission and evidence reuse

Figures 7 and 8 show image transport, retained evidence and bounded capture ownership.

Some observations can be omitted before native work. A direct byte comparison can establish that the original and transformed inputs are identical. ZIP metadata alone does not authorize that omission. A separate rebuild-eligibility check reuses dependency facts from the package scan and a strict part/feature allowlist. Formulas, refresh sources, unknown extensions, duplicate members and unresolved metadata keep rebuilding enabled. Cached values or an empty capture plan do not establish calculation independence.

Completed-result reuse requires compatible source identities, selected parts, implementation, state-preparation policy and rendering environment. Uncompressed member digests make identity independent of ZIP order and compression. A prior part selection must cover every newly requested component. User-selected weaker component policies remain explicit evaluation choices, with correspondingly bounded claims. Interrupted runs and missing image receipts cannot supply completed evidence.

IV. Evaluation

RQ1 asks whether the selector captures the known visible effects in the tested mutation classes and independent reference sample. RQ2 measures how much native capture work and elapsed time it avoids. RQ3 examines the reproducibility and investigability of retained findings, including failures and rendering variation.

A. Corpus identity and outcome accounting

The evaluation treats Data.gov, Destatis and Enron as distinct populations. Archive checksums, member names and duplicate occurrences identify each selected corpus version. Legacy XLS conversion is reported separately from XLSX cleanup. Recursive extraction of the EDRM Enron v2 XML archives from Internet Archive inspected nested archives and embedded containers. The resulting handover contains 34,065 distinct candidate SHA-256 identities totaling 8.470 decimal GB of uncompressed payload. Manifest checks verified transport while retaining original bytes and upstream extraction provenance. The frame includes legacy Excel and other OLE candidates. Native format recognition, XLSX conversion and visual eligibility are separate outcomes, so the candidate count does not measure workbook coverage.

The frozen source inventories contain 9,117 Data.gov XLSX files totaling 25.273 GB and 1,182 Destatis XLSX files totaling 2.364 GB. Each listed archive member has a SHA-256 identity. These source-frame sizes are separate from the eligible, completed and independently referenced subsets.

B. Sampling and reference controls

Seeded samples use frozen source frames and size strata chosen before visual inspection. Development stress cases are labelled separately. Independent reference plans bypass candidate selection. Timeouts and unsupported inputs remain in

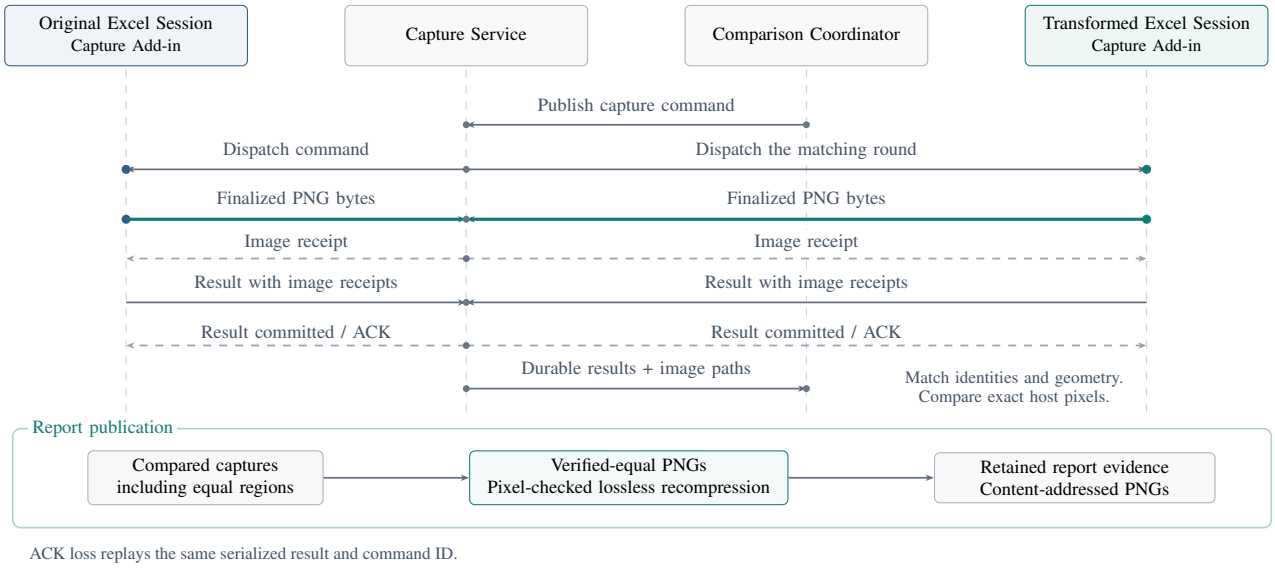


Fig. 7. Image transport and retained evidence. Each Excel session contains its workbook and capture add-in. Finalized PNGs receive image receipts before completed commands are acknowledged. Durable results bind the comparison partners and geometry. Equal captures remain inspectable in the report. Lossless recompression is accepted after decoded-pixel equality is verified. A lost acknowledgment replays the same result and command identity.

the selected denominator. Controlled edits require an observed native effect or an explicit invariance outcome.

A reconciled historical Enron development sample retained twelve seeded source occurrences. Seven completed native references over 33 declared rectangles, two failed image-size preflight and three required unsupported graphical or embedded-object envelopes. All 33 completed rectangles had identical overlapping RGBA pixels. Two had additional opaque-white exterior strips, totaling 196,657 pixels, which were retained as raw extent differences. This sample is not held out, and failed or unsupported references remain in its denominator.

A separate sample selected one input per size quartile from 30,301 prepared Enron XLSX files using a fixed seed before inspecting selected outputs. All four native references qualified over 41 stored-dimension rectangles, with matching image dimensions and zero changed RGBA pixels. Prior development exposure remains unclassified. The ordinary v15 selector rendered 14 images totaling 78.0 million pixels, compared with 82 reference images totaling 398.0 million pixels. All seven targeted image pairs were equal. One byte-identical input pair required no targeted images. These are render-work counts under distinct plans, with adaptive targeted partitions and fixed 256-row reference tiles [6].

A fixed-seed Data.gov sample selected one member per size quartile from the 9,117-entry inventory before joining cleanup outcomes. Two initial references failed size preflight. Smaller independent partitions subsequently completed those inputs, retaining the failed attempts. All four references cover 188 areas with equal dimensions and pixels. Targeted selection used eleven equal pairs. Prior exposure is unclassified; this all-equal sample cannot estimate recall on changed inputs [7], [8].

A separate twelve-occurrence Destatis sample used the same fixed-seed, size-quartile selection policy. All twelve selected packages contain graphical or embedded parts and remain unsupported by the current dimension-only reference preparer. They were not replaced by easier inputs. An optional native-object qualification extension admitted one of the same twelve inputs over two fixed areas. Its image rectangle fit inside the cover-sheet area, and both reference areas had identical overlapping RGBA pixels. The data-sheet image had a 20-pixel-high opaque-white exterior strip totaling 25,820 pixels. The remaining eleven inputs retained unsupported outcomes for legacy drawings, embedded objects or drawing effects. This extension checks native object rectangles, with explicit rejection of unsupported types and out-of-area objects. It does not certify decorative ink beyond those rectangles or establish corpus recall.

C. Detection and cost metrics

A completed independent reference determines whether a mutation has a visible effect under the recorded observation policy. A selector miss occurs when such an effect falls outside selected coverage. Detection is measured per mutated workbook pair and per mutation class. A partially covered effect retains a separate outcome even when another difference makes the pair detectable. Pixel coverage requires a documented mapping between reference and selected capture coordinates, including extent differences.

Selection overhead counts candidates whose completed paired images are equal. This measures avoidable work within the tested policy. Pixel findings retain their attribution as transformation effects, rendering variation or unresolved differences. Failed references, missing partner images and timeouts retain explicit outcomes. Report both completed-reference

TABLE II
DESIGNED MUTATIONS AFTER NATIVE REBUILD. PIXELS COUNT CHANGED RGBA LOCATIONS IN INDEPENDENTLY FIXED REFERENCES. EVERY SELECTED MASK EXACTLY MATCHED ITS REFERENCE MASK AT THEIR COMMON A1 ORIGIN. THESE CONTROLS DO NOT ESTIMATE POPULATION RECALL [9], [4], [10], [11].

Mutation	Pairs	Reference pixels	Selected area
Wrapping	1	1,775	A1:D3
Merge A1:D1	1	3,873	A1:D3
INDIRECT conditional fill	2	5,070 each	A1:AA3
Formula errors	2	366 / 358	A1:D3
Color scale	1	5,727	A1:D3
Data bar	1	4,304	A1:D3
Defined-name binding	1	53	A1:D3

detection and the fraction of all selected cases with usable reference evidence.

D. Mutation strata and statistical scope

Table II summarizes nine designed controls with exact reference-mask coverage. INDIRECT used the unsupported-rule fallback. Dependency resolution and broader unresolved-dependency strata remain open. The defined-name case changes only the range binding, rebuilding its unchanged SUM formula from 3 to 6. The color-scale and data-bar cases change rule colors.

Both states are rebuilt and saved. NOW, RAND, INDIRECT and OFFSET require reference-policy controls [12]. Raw pixels and calculation context support attribution, alongside identical-input rendering controls.

For zero misses in n independent trials with common miss probability, the exact one-sided 95% upper bound is $1 - 0.05^{1/n}$ [13]. Workbook mutations share a cluster. Population estimates require strata and inclusion probabilities, with development cases reported separately.

E. Matched cost accounting

Cost accounting separates phases, images, pixels, retained bytes, elapsed time and coverage unions. Overlapping captures count as repeated work. Matched trials control preparation, environment, image budgets and reuse through open, rebuild and save. Repeated pairs measure variation. Timeouts remain censored. Corpus projections require geometry-calibrated strata and hidden-region/object-extent assumptions. Production queues cannot isolate tuning effects.

F. Streaming cell-comparison cost

The Figure 3 implementation completed 63 fresh-process trials on an AMD Ryzen 7 7800X3D with sixteen logical processors, Windows 11 and 102,066,212,864 OS-reported physical bytes [14]. The release executable uses rustc 1.96.0 and LLVM 22.1.2. All expected mutation counts passed, as did 27 focused correctness tests, including 6,561 ordered comparisons against an independent coordinate/value oracle. For one million densely changed cells, immediate consumption and retained changes had maximum process working sets of 5.703 and 94.973 MiB, respectively, a 94.0% reduction for immediate consumption. Median combined stream-construction times were 671.849 and 755.226 ms, with median merge times of 66.536 and 94.204 ms. Both wrote 80,000,000 scratch bytes.

The three real worksheets emitted exactly one change per fixed single-value mutation. The largest 6,486,777-cell pair had median construction and merge times of 4,886.604 and 393.002 ms, 518,942,160 scratch bytes and a 5.934 MiB maximum working set. Across the real-input trials, one repetition of a 3,060,845-cell case peaked at 42.121 MiB working set. Its other two repetitions peaked at 6.105 and 6.078 MiB. The higher peak remains in the receipts and maxima without a causal attribution. Working set and private commit retain their separate counter semantics. Native comparisons continued concurrently and normal filesystem caching remained enabled. Input preparation and decompression remain outside the timed component, and full-pipeline evidence retention is outside this result-retention comparison.

G. Opening, batching and independent Excel pairs

A bounded opening experiment used four copies of a synthetic workbook with eight million-row, three-column sheets on one Windows host. Two reversed-order rounds measured median batch times of 118.388 seconds sequentially and 41.714 seconds in independent processes sharing a private desktop [15]. The endpoint includes Office.js readiness. Staging and teardown are excluded, and cache and background-load effects remain. This supports overlapping startup for that configuration, not arbitrary concurrent COM calls.

A warmed-up 64-range geometry experiment returned identical measurements in scalar and batched modes. The scalar passes each took 16.094 seconds and the batch passes 0.250 seconds. Twelve-region image packages, including four zero-area regions, took 3.125 seconds at requested size one and 0.579 seconds at size eight, with identical decoded image dimensions and pixels. Opening was excluded and 250-ms observation polling quantizes these timings [16]. The experiment establishes a transport-amortization opportunity under its recorded workload rather than a corpus throughput factor.

A paired experiment used one host and shared inactive desktop with independently owned pair states. Over three repetitions per completed configuration, moving from one pair

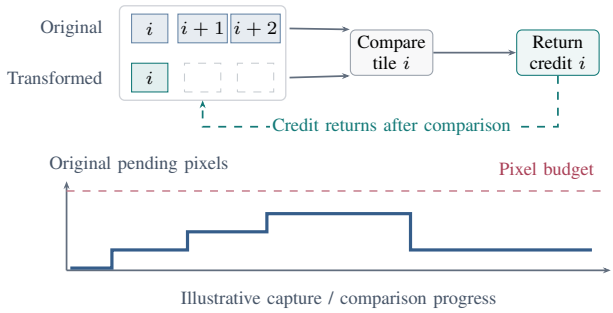


Fig. 8. Bounded capture ownership. Credits remain reserved from issue through paired comparison, so a faster client retains bounded pending work while its partner catches up. PNG buffers also have byte limits and explicit lifetimes. The occupancy trace is illustrative.

to two increased capture-only throughput from 3.752 to 7.030 pairs/minute with binary PNG transport and from 7.590 to 12.312 with hash-first transport. Peak observed job memory rose from 1,525 to 2,685 MiB and from 1,472 to 2,666 MiB, respectively. Every tile in this selection was equal, favoring the hash-first path. Requested configurations for three through eight pairs were rejected before launch by RAM admission. Those rejections are not zero-throughput measurements or an Excel scaling ceiling [17], [18].

H. Coverage qualification before speed

A synthetic million-row request returned an Office.js PNG without an API error in 2.58 seconds. It contained only $85 \times 32{,}766$ pixels against the required 20,971,521-pixel height. The boundary probe found 1,638 rows complete and 1,639 incomplete on that installation [19]. This demonstrates why completion and dimensional qualification are separate conditions.

The historical 62-case protocol completed 48 cases and retained 14 failures. Four unresolved drawing-bound worksheet outcomes account for repeated failures. They are not 14 independent crashes. Six of eight coverage workbooks completed, covering 462 of 466 requested worksheets. The failed observations remained incomplete instead of receiving a smaller undocumented range or an equality classification [20].

I. Reducing work while retaining the selected union

The FRPP workbook has 307,920 rows, including 304,140 hidden rows. Of 19,248 original grid requests, 18,872 have zero area. A read-only geometry probe drove the Rust adaptive planner to 64 leaves after 710 probes in 2.815 seconds. This is a planner observation, not a completed paired Office.js comparison. A separate native visibility-transition control exposed a row only in the transformed state. It retained the original's zero-area observation and reported rendering and geometry differences without claiming paired-pixel comparison there [21].

A 208-worksheet diagnostic exposed 5,192,711 intermediate candidates. Grouping contiguous changed row formats reduced that count to 603,267 with identical content envelopes. Both diagnostics stopped before complete rasterization. This establishes reduced intermediate representation size and no measured end-to-end speedup.

Uniform inherited backgrounds exposed a related native-work amplification. One Destatis pair had 128,912 versus 2,197 column definitions. The former plan measured 16,101 regions, although most were outside the explicit-cell envelopes. The guarded finite-background plan produced 165 image pairs, all identical in a fresh comparison. Selection and transport took 41.371 seconds and comparison 1.379 seconds. The earlier attempt was interrupted after more than 40 minutes. A second Enron pair reduced a former 65,664-region plan to 73 measured regions, all with identical images. Its selection and transport took 28.114 seconds. The latter case retained an empty finite AutoFilter and all serialized-cell extent. These are selected development observations with censored earlier attempts, not matched speedup estimates. Regressions retain distant fill transitions, borders, overrides and unsupported rule contexts.

Among 134 retained SheetShrink comparisons against the selected originals, median selection and transport took 7.4 seconds on Destatis (42 pairs), 30.4 on Data.gov (46), and 3.0 on Enron (46). Historical capture configurations vary. These timings describe recorded work, without estimating a speedup against full capture.

An inherited border can extend past every populated cell. In the Data.gov QT/RR workbook, the rebuilt dimensions of eleven affected sheets end at column L, but Inquire removes an inherited bottom border on row 1. An explicit reference at `XFB1:XFD3` found a 193-pixel line missing at the far edge, with equal measured geometry. Of 90,112 rectangles in the ordinary capture plan, 2,101 intersect columns A:L or row 1. This count describes a potential reduction in capture work. The two regions alone do not qualify a smaller plan because other layout and inherited appearance changes may remain.

Conversely, unsupported objects can dominate a conservative plan. Two retained attempts contained explanatory sheets responsible for 104,833 and 185,856 regions. After approximately 5.3 hours, neither full capture had completed. Removing two zero-height lines in a separately supplied reference area changed 1,042 RGBA pixels because their strokes remained visible. The long-running fallback therefore cannot be removed merely because an object's geometric height is zero.

J. Comparator pilot with independent native controls

Five frozen synthetic pairs were supplied to xlCompare 12.5.0.0 and DiffEngineX 3.18.0.0 through their report commands. Both exports included an offscreen B2000 edit. Neither listed a hidden B3000 edit under the tested commands. Direct visual and sparse-formatting changes produced no exported cell differences. A dependent pair exported the changed B1 precedent without a separate conditional-fill or chart-rendering finding. Edition and option coverage remains incomplete, including xlCompare's reported first-ten-rows export limit.

DeltaSieve completed the same five pairs using their ten hash-verified inputs. The identical pair was excluded by byte identity. Four pairs produced native pixel findings. The offscreen edit changed 47 overlapping pixels. Direct, dependent and sparse-formatting pairs changed 7,057, 3,950 and 1,281 grid pixels, respectively. The dependent pair also changed 36,505 chart pixels. The region containing hidden B3000 rendered equally.

An independently supplied reference plan bypassed candidate selection and content clipping. It used Data!A1:N30 for the short pairs, and continuous 256-row tiles through rows 3072 and 2304 for offscreen and sparse pairs. All five comparisons completed. The identical control remained equal. Only the B2000 and J2099 tiles changed in their respective long pairs, while the direct and dependent pairs also differed in their broader reference areas. Pair-level outcomes agree with the selected run. This is a bounded control, not an estimate

TABLE III
NATIVE CHART CONTROLS. IMAGE COUNT INCLUDES BOTH STATES. AN EXCLUDED CHART SUPPLIES NO NEW PAIRED-PIXEL OBSERVATION.

Control	Images	Exclusions	Differences
Unchanged, deterministic	0	1	0
Unchanged, conservative	2	0	0
Changed source, deterministic	2	0	1

of corpus recall or pixel-union coverage. Different chart and capture extents prevent pooling their raw pixel counts. No matched cross-tool speedup follows from these trials.

Two representation controls permuted cell-XF and shared-string indices while remapping their cell references. Independent `Data!A1:N30` renders were pixel-identical within both pairs. The style permutation selected zero regions. The shared-string permutation retained one equal image pair at `A1:D4`, exposing unnecessary capture work under the tested policy v15.

The wrapping and merge controls in Table II use independent `A1:N30` references. Merged A1:D1 includes populated cells and exercises native merge behavior [9].

K. Chart dependency controls

A workbook created in Excel supplies three controlled chart cases. The unchanged pair differs only in document-description metadata. Deterministic selection excludes its one chart, while conservative mode renders both partners. A third pair changes a source value and retains chart capture. Table III records the observed outcomes [22].

These controls test both sides of the exclusion gate. Equal caches alone are not the sufficient condition. The package and source-cell dependencies and native geometry must also agree. A separate rebuilt Lysimeter inventory contains 69 charts. Its gate excludes 34 and retains 35. Eighteen fallbacks have ambiguous duplicate names and 17 reference additional user-shape dependencies. This inventory did not complete a replacement image comparison and supplies no end-to-end runtime reduction.

L. Attribution controls and retained counterexamples

Independent controls distinguish a transformation defect from capture or renderer behavior. In a fixed-height statistics sheet, the original-versus-original reference was pixel-identical, while the transformed pair repeatedly changed 25,640 pixels. Restoring 87 removed font-bearing blanks recovered the original image hash. A corrected SheetShrink output also matched over the frozen A1:L916 reference area. The case demonstrates a real witness omission rather than an excuse to loosen pixel comparison.

A migration control instead exposed inconsistent exterior context. Its relevant content ended at row 84, but capture included an empty row 85. That row changed from 12.75 points to hidden. A zero-width region then produced a height finding without an image. Applying the same union-content boundary to geometry and capture removed this exterior-context finding. A fresh run completed 38 regions, including 37 pixel comparisons, with no capture-region difference there. Separate height observations remain recorded [23].

Five Enron controls differed only in their ZIP comments before opening. Conservative capture compared ten charts. Eight were pixel-identical. Two charts, previously flagged in the SheetShrink comparisons, showed two and seven changed pixels in these controls with equal geometry. Rebuilt worksheet, style and chart parts were unchanged between control lanes. The controls demonstrate that native rendering can vary in these charts without a workbook-member transformation. Attribution of the original pixel masks remains unresolved, and the report retains the original findings alongside the separate controls.

Embedded-object mutations test another boundary. Moving a PDF-icon frame changed 6,473 RGBA pixels in a fixed reference area. Changing only its cached preview changed 5,242. Separately enlarging three zero-size text shapes produced 1,531, 1,737 and 1,487 changed pixels on three sheets, while removal controls were pixel-identical in their supplied areas. These cases show why package anchors, previews and visible ink need distinct treatment. They do not establish coverage for arbitrary OLE formats.

M. What the measurements establish

The ledger separates package bytes, candidates, native rectangles, rendered pixels and compared pixels. Candidate coalescing, adaptive partitioning and hash-first transport affect different stages and retain separate denominators.

Similarly, process memory is not a single interchangeable counter. Native Excel, WebView2, the local host and the analysis process can hold different buffers. A sampled private-byte maximum can miss a short allocation peak, and a working-set maximum is not private commit. Scope and sampling interval accompany the reported memory. Phase durations can overlap and are not added as if they were independent. A claim about total campaign throughput includes startup, preparation and failed attempt costs under a declared reuse policy.

The independent reference plan remains distinct from those optimizations. It renders externally supplied areas without candidate selection, conditional-rule exclusion or production content clipping. Its author is responsible for the asserted coverage of those areas. A failed or oversized reference remains a failed reference case. Agreement between a selected plan and a reference over five deliberately constructed pairs supports those controls. It does not turn the same five pairs into a representative sample of three corpora.

N. Recorded comparison coverage

The retained snapshot covers the nine workbooks with the largest absolute SheetShrink file-size reductions per corpus in the frozen selection. Each workbook contributes one slot per tool. Table IV reports a timestamped snapshot of all 135 slots. The versioned snapshot identifies each rank and tool [24]. The separate Enron ledger retains the detailed outcomes and baseline identities [25].

TABLE IV
RETAINED COVERAGE SNAPSHOT AT 00:50 CET ON 11 NOVEMBER 2026. EACH CORPUS HAS 45 SLOTS. RAW AND NORMALIZED COUNT COMPLETED COMPARISONS WITH LIMITED COVERAGE. UNAVAILABLE COUNTS MISSING VARIANTS. NO ELIGIBLE COUNTS SLOTS WITHOUT A QUALIFYING WHOLE-WORKBOOK RESULT.

Corpus	Raw	Norm.	Unavail.	Failed	No eligible	Active
Destatis	27	2	1	14	1	0
Data.gov	38	0	0	4	1	2
Enron	37	1	4	3	0	0

Of the three Enron failures, two followed Excel repair prompts and one required unresolved whole-grid coverage. The repaired Check Performance preparation and the completed SheetShrink comparison at rank two use different baseline hashes. The repaired preparation therefore provides no matched visual observation for that tool slot. These counts describe coverage of the selected high-reduction workbooks. Detection recall requires the independent reference experiments.

O. A border difference inside the common capture

An Enron case study rendered `Tables!A1:I157` after inspecting a Check Performance finding. Four independent original captures were pixel-identical. SheetShrink and XLStylesTool preserved that reference exactly. Check Performance differed at 126,404 common pixels, while Inquire differed at 490 near the lower edge. Extending the reference through row 160 retained 127,256 and 916 changed common pixels, respectively. Inspection of the extended Inquire capture showed a missing lower horizontal table border inside the shared rectangle. Its rebuilt worksheet lacked original row 157 and its styled blank cells. The added context therefore separates this finding from a capture clipped at the table edge. NXPowerLite had no available output for this input. This post-hoc illustration covers one declared area and does not measure recall or establish whole-workbook preservation.

a) Different capture coverage for the same original: For one Data.gov PFAS workbook, the original package-member inventory matched across three tool comparisons. Inquire yielded 129 differing regions among 4,277 selected regions. An inspected range contained purple original pixels that were white in the candidate inside the common image extent. Check Performance yielded no differences in its 63 regions. XLStylesTool selected zero regions. These observations have different denominators and cannot rank the tools' visual preservation.

Figure 1 shows the same `readme!A1:M22` reference area across all five tools. The original images agree across observations. Check Performance, Inquire and NXPowerLite omit the fill of D2:L2, changing the same 16,182 pixels. The two strips use the same crop coordinates. The reference manifest records source and image hashes [26]. This localized result supplies no recall estimate.

P. Geometry and finite background coverage

a) Merge order and effective row height: In Data.gov's `CyanoPhyto_qPCR_dataset_87a55675e15a.xlsx`, the Original–Check Performance comparison selected the full grid `A1:XFD1048576` on `qPCR_Data_Second`. Its 17 merge rectangles were merely reordered, but a row border-height hint also changed. Canonicalizing the merges alone retained the fallback. Native height checking measured row 341 changing from 16.5 to 15.75 points, yielding the logical layout interval `A341:XFD445`. Other cell, conditional-formatting and chart obligations remained selected. The revised workbook comparison completed 525 cell regions and 49 charts per state in 245.2 seconds, with differences retained for inspection [27].

The earlier full-grid attempt did not complete. For an explicitly conditional cost projection, repeating an observed 32×256 tail tile over the full grid in both states gives $2(16384/32)(1048576/256) = 4{,}194{,}304$ images. At the recovery's measured capture-selection-and-transport rate of 6.76 images/s, this uniform-tile, constant-throughput scenario takes approximately 7.2 days. Adaptive partitioning, image entropy, caching and concurrency can change that cost. The projection and the measured 245.2-second end-to-end recovery have different endpoints and supply no measured speedup ratio.

b) Finite backgrounds with row borders: A designed fixed-default-height change previously produced an unbounded A1:XFD1048576 plan and stopped at the guard. Policy v17 completed 69 regions from A1:XFD4, retaining the far-right row border. Aligned change masks matched independent A1:D5 and XFA1:XFD5 references at 1,170 and 674 common pixels. An initial control without customHeight rebuilt to the same 15-point default in both states, yielding equal references and zero selected regions. It remains an invariance outcome [5].

Nine zero-width `LCCA` ranges differed in measured height (2000.25 versus 1827 points) without images. Version 16 retains their geometry but excludes these changes from visual findings after validating both zero surfaces. Missing images and visibility transitions remain independently checked. An unchanged-input native repeat completed 29 worksheet and 27 chart observations in 216 seconds with zero visual differences and nine nonvisual geometry observations. Coverage remains limited to the selected areas.

Q. Evaluation limits

Bounded controls, stress cases and partial campaigns do not establish population recall, corpus-wide equivalence or matched end-to-end speedup. Broader independent references, mutation coverage and common-endpoint cost trials remain open. Manifests retain identities, revisions, settings and repeated attempts, with distinct complete, different, unsupported, failed and unfinished denominators.

V. Related Work

A. Spreadsheet change inference

SheetDiff infers changes between spreadsheet versions and presents them in a single contextual view [28]. RowColAlign extends longest-common-subsequence reasoning to two-dimensional alignment and evaluates edit recovery using generated pairs with planted changes [29]. Its known edit history provides a methodological precedent for controlled mutations. Our evaluation additionally requires native observations of which mutations produce visible effects under the chosen state-preparation policy. Edit recovery and rendered-effect coverage have different observation units.

B. Spreadsheet dependencies and testing

Spreadsheet slicing connects dependency analysis with integrated testing and fault localization [1]. DeltaSieve uses supported dependencies to select native observation regions. Fisher et al. retain spreadsheet test cases and use change-impact analysis to support their reuse [30]. TACO compresses formula graphs using regularity in cell references and supports queries over the compressed representation [31]. DeltaSieve follows supported conditional-formatting and chart inputs to affected presentation regions. Its coverage obligations include inherited formatting, automatic row geometry and object bounds alongside formula dependencies.

C. Workbook comparison and native evidence

Spreadsheet Compare documents workbook comparison reports with formula and cell-format categories, including hidden sheets [2]. Its documented comparison categories establish formatting-aware spreadsheet differencing. DeltaSieve evaluates native rendered effects and records the selected capture scope. The documentation describes change reports and a side-by-side grid, while pairwise native image validation of reported candidates requires evidence from the tested product and configuration.

xlCompare documents command-line comparison and exported difference reports [32]. DiffEngineX documents whole-workbook and folder comparisons, report workbooks, and comparisons of formulas and constants. Its help explicitly excludes chart comparison [33]. xlCompare also documents configurable formatting comparisons [34]. These capabilities support the shared-input pilot with exported reports. A report of edited cell inputs and an observation of their dependent rendered consequences answer different questions. The pilot must therefore record the tested options and report coverage instead of treating every comparator as an interchangeable visual oracle.

Synkronizer's edition documentation lists content, formatting and Excel-name comparison features [35]. Spreadsheets-Diff advertises two- and three-way comparison, key-column matching, and workbook/VBA comparison features [36]. A native pixel-observation contract and its tested coverage require separate verification for each product and version.

D. Cross-system testing and parsing

SQuaLity examines barriers to reusing database test suites across systems, including configuration and execution dependencies [37]. This suggests recording the renderer and environment when reusing observations. The analogy does not transfer database test adequacy guarantees to spreadsheet rendering. SheetReader's specialized parsing work [38] is relevant to efficient package inspection, whose cost must be separated from native rendering cost.

E. Selective regression testing and visual models

Ekstazi selects regression tests through recorded file dependencies [39]. It tracks dynamically used files and their checksums. Its separation of analysis, execution and dependency collection motivates measuring selection overhead together with avoided work. DeltaSieve's spreadsheet dependency model and native image observations require their own coverage evaluation.

ReDeCheck detects potential responsive web layout failures from structural models [40]. VISER renders reported issues in a browser and manipulates element opacity to determine whether an effect is visible [41]. VERVE extends this approach to all five ReDeCheck failure classes and uses histogram-based image comparison [42]. Structural filtering followed by targeted visual verification is therefore an established method. DeltaSieve applies this principle to paired workbook states and spreadsheet presentation dependencies, including effective formatting, conditional rules, chart sources and native geometry. Its coverage evaluation must independently assess which affected regions the selector retains.

F. Visual document regression and benchmark scope

Huang's account of the Antenna House Regression Testing System (AHRTS) describes rasterizing baseline and new PDFs and comparing their page images pixel by pixel [43]. Document and image checksums avoid work before more detailed comparison. This provides a document-processing precedent for staged visual regression testing and reuse. DeltaSieve selects regions within workbooks using spreadsheet dependencies and retains native geometry alongside pixel observations.

The evaluated scope concerns XLSX transformations observed in Excel. VISER and VERVE consume web pages and DOM-based layout reports. Adapting their selectors would require a spreadsheet-specific dependency and geometry model. Applying AHRTS through PDF export would introduce print areas, pagination and scaling into the observation contract. These input and observation differences motivate methodological comparison with these systems. Direct runtime rankings would require a common task, preparation policy and capture scope.

VI. Discussion and Limitations

The central risk is an incomplete candidate model. A proof for supported inputs must state its dependencies and applicable namespace or rule subset. Unknown features need a capture fallback or an explicitly incomplete outcome. Runtime limits

and failures reduce achieved coverage and must remain in the report.

A native renderer is also an experimental dependency. Recalculation, volatile formulas, file-location functions and rendering variability can change output without a transformation defect.

Current reports incompletely record whole-sheet fallback scope and per-candidate direct or indirect XML provenance. Attribution requires separate review where this information is unavailable.

VII. Conclusion

DeltaSieve combines dependency-aware region selection with native rendering and inspectable evidence. The bounded experiments demonstrate concrete findings and fallbacks. Corpus-wide recall and matched capture cost remain open.

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The author directed development, reviewed source code, and performed manual testing and debugging. Agent observations supported investigation. Reported validation outcomes are grounded in the retained native captures and comparison records. The author remains responsible for the scientific claims, references, evidence selection, originality and final submitted content.

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